Constructors in Java

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What is a constructor in Java?

A constructor in Java is a block of code like the method. It is called when an instance of a class is created. At the time of calling constructor, memory for the object is allocated in the memory. It is a special type of method which is used to initialize the object.

Every time when we. Create an object by using new keyword. It calls a default constructor. If there is no constructor available in the class. In such case Java compiler provides a default constructor by default.

Note: It is called constructor because it constructs the values at the time of object creation. It is not necessary to write a constructor for a class. Because Java compiler creates a default constructor if your class does not have any.

Rules for creating Java constructor

1. **Constructor name must be same as its class name:** A constructor must have the same name as the class. This helps the Java compiler identify the constructor and use it while creating an object.
   For example:

```
package ClassAndObjects;


/**
 * Write a description of class Person here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Person
{
    private String name, mobileNum;
    int x;
    float f;
    double d;
    //default constructor initializes the instance variables
    // with default values, Integer have 0, String will be null
    // float => 0.0f, double => 0.0
    public Person(){
        System.out.print("\nConstructor called\n");
    }
```

```
}
```

2. **A constructor must not have no explicit return type:** A constructor does not return any value, not even void. If you specify a return type, it will be treated as a normal method, not a constructor.

3. **A Java constructor cannot be abstract, static, final, or synchronized. (will be explained later)**